

Teen Numbers: One Ten and Some Ones

Answer Key

Kindergarten / Grade 1 Number and Operations in Base Ten

Quick Answers

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|-------|------|---------------|---------|
| 1. 13 | 3. 5 | 5. — | 7. 7 |
| 2. 16 | 4. 8 | 6. 12, 16, 19 | 8. —, 4 |
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Curriculum

Level: Kindergarten / Grade 1 Number and Operations in Base Ten

1.NBT / 2.NBT.A – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–4 of 5 across 9 tagged checks.

Common wrong answers

3. *If they got 15: counted the whole number as ones and wrote the teen number itself in the ones box*

1. Two ten-frames: one full, one with three counters.

Step 1. The first frame is completely full. A full ten-frame always holds ten — the child should recognise it without counting the dots one by one, which is the whole point of the frame.

Step 2. The second frame has three counters in it, so the ones are 3

Step 3. Put the two parts together: $10 + 3 =$ 13

Listen for: a child who recounts the full frame from one. Ask “how many does a full frame hold?” before they start counting.

2. One ten-stick and six loose cubes.

Step 1. The stick is made of ten cubes joined together, so it counts as one ten. It does not need counting.

Step 2. Count the loose cubes: there are 6

Step 3. $10 + 6 =$ 16

Note: this is the same idea as problem 1 in a different material. Say the sentence aloud both ways — “ten and six” and “sixteen” — so the number word and the model line up.

3. Draw 15; then 15 is one ten and how many ones?

Step 1. Fill the first frame all the way up. That uses ten of the fifteen counters.

Step 2. Fifteen take away ten leaves 5, so five counters go in the second frame — one row of five, which is easy to see.

Step 3. The sentence reads: 15 is one ten and 5 ones.

Common wrong answer: 15 in the ones box — the whole number was copied instead of just the leftovers. The first frame is already used up, so the ones can never be more than nine.

4. Draw 18; then 18 is one ten and how many ones?

Step 1. First frame full again: ten counters.

Step 2. Eighteen take away ten leaves $\boxed{8}$, so the second frame gets eight — a full top row of five and three underneath.

Step 3. 18 is one ten and 8 ones.

Note: the tens digit of every teen number is 1, and the ones digit is exactly the number of leftover counters. Point at both while saying it.

5. Compare the two piles.

Step 1. Both piles have exactly one ten-stick, so the ten does not help us decide. Cover the sticks with a hand and look only at the loose cubes.

Step 2. The left pile has four loose cubes and the right pile has seven. Four is less than seven.

Step 3. $\boxed{10 + 4 < 10 + 7}$ — that is, fourteen is less than seventeen.

Listen for: the mouth of the symbol opens toward the bigger pile.

6. Order 16, 12 and 19 on the number line.

Step 1. Every one of these is one ten and some ones, so they all sit between ten and twenty on the line. Only the ones digit moves them.

Step 2. Two ones, six ones, nine ones — reading the line from left to right puts them in that order.

Step 3. Smallest first: $\boxed{12, 16, 19}$

Note: the number line is doing the same job the ten-frames did. Ask which one is nearest twenty and why.

7. Number bond: the whole is 17, one part is the ten.

Step 1. The circle on top holds the whole number, and the two circles underneath hold its parts. One part is already filled in as the ten.

Step 2. We need the number that goes with ten to make seventeen. Count on from ten: eleven, twelve, thirteen, fourteen, fifteen, sixteen, seventeen — that is $\boxed{7}$ counts.

Step 3. $10 + 7 = 17$, so the missing part is 7.

Note: this is the reverse of problems 1 and 2. There the parts were given and the whole was missing; here the whole is given and a part is missing. Same bond, read the other way.

8. Ella's pile against Mo's pile.

Step 1. Both children have one ten-stick, so again only the loose cubes matter. Ella has nine, Mo has five.

Step 2 (part a). Nine is more than five, so $\boxed{10 + 9 > 10 + 5}$ — nineteen is greater than fifteen.

Step 3 (part b). "How many more" is a subtraction. Because the tens match, subtract the ones or count up from Mo's number to Ella's: fifteen, sixteen, seventeen, eighteen, nineteen. Either way the answer is $\boxed{4}$

Note: the two tens cancel out. A child who tries to subtract nineteen minus fifteen by counting all the cubes will get the same four — slower, but correct. Show them the shortcut afterwards, not instead.